

What is a Psychopathology Dimension?

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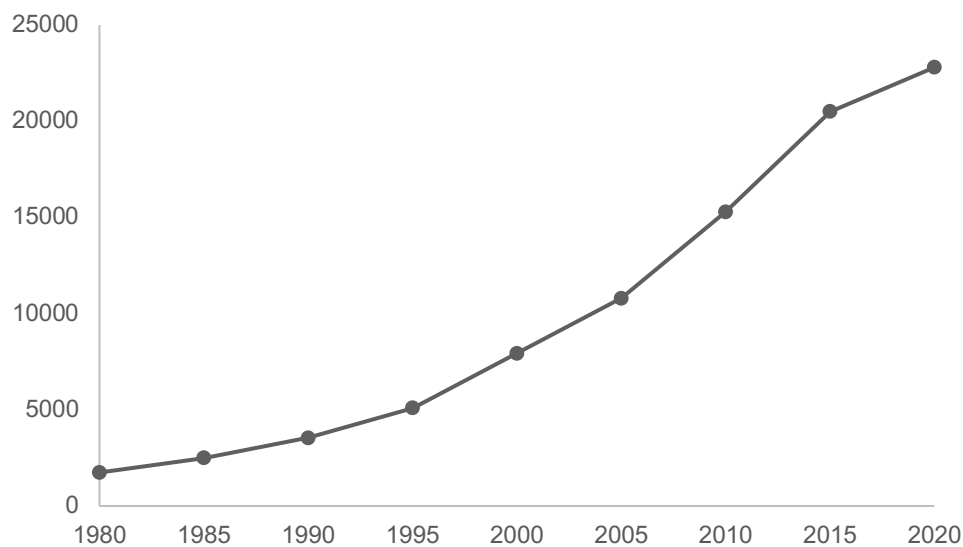
Abstract

Coherence in the science and practice of mental health assessment depends upon a tight connection between psychopathology concepts that are used and the way those concepts are operationalized and defined. In contrast, the use of the same word to mean more than one thing contributes to incoherence, inefficiency, and confusion. In this paper, we review three possible meanings of the word “dimension” as it relates to the assessment of psychopathology and describe how the indiscriminate use of this word has caused confusion in the general context of the transition to a more evidence-based approach to mental health diagnosis. We attempt to disambiguate the term “dimension” by demarcating three concepts that can be distinguished based on different empirical standards: continuous variables, unidimensional dimensions, and distinct dimensions.

Keywords: Psychopathology, Assessment, Measurement, Dimension, Category, Psychometrics

Psychopathology research and clinical practice is undergoing a transition from categorical to dimensional models of mental health diagnosis. The number of papers about psychopathology dimensions has increased dramatically during the last few decades (Figure 1). Many authors have explicitly advocated for “dimensional models” (e.g., Clark et al., 1994; Harkness et al., 2014; Haslam et al., 2020; Hopwood et al., 2018; Krueger et al., 2005, 2018; Ofrat et al., 2018; Patrick & Hajcak, 2016; Watson et al., 2008; Widiger, 1993, 2005; Widiger & Simonsen, 2005; Widiger & Trull, 2007). It has become so popular to support the transition from categorical to dimensional models that the main opposition generally has to do with recommendations to do so cautiously and thoughtfully, rather than not to do so at all (e.g., First, 2005; Herpertz et al., 2017; Huprich & Bornstein, 2007; Jablensky, 2018; Zachar & First, 2015).

Figure 1. *Google Scholar hits for “psychopathology dimension” from 1980 to 2020 (retrieved February, 2022).*



Generally speaking, the difference between categorical and dimensional models is that whereas categories distinguish qualitatively between people who have a disorder and people who do not, dimensional models distinguish these groups in terms of quantitative

gradations. However, in psychopathology research and taxonomic debates, the term “dimensional” has been used to indicate that a variable meets different kinds of criteria: a) that its indicators can be represented with continuous or quasicontinuous variables (for example, a sum or average), b) that its indicators fit a superordinate factor model of some sort (for example, a one-factor model, or a hierarchical model with one dominant factor), or c) that the variable is homogeneous and distinct from other, potentially similar dimensions.

In the first section of this paper, we describe how using the word “dimension” to refer to different concepts has muddled diagnostic debates. We then describe how different kinds of empirical criteria can be used distinguish variants of dimension from one another. We conclude with a discussion of some adjacent issues relevant to the use of the term dimension in the mental health literature.

The Problem

Since the beginning of modern psychological assessment, various authors have emphasized for the need to clearly distinguish psychometric concepts, for the purpose of a sharp, structured, and methodical assessment science (Cronbach & Meehl, 1955; Loevinger, 1957; Messick, 1988; Slaney & Maraun, 2008). We focus in this paper on one element of this larger need: the use of the word dimension to describe psychopathology constructs in the context of transitions away from diagnosis based on polythetic disorder categories.

The problem in this context is that the term dimension has been used inconsistently in discussions about the relative advantages of “dimensional” and “categorical” models or constructs. For example, some authors have used the word dimension to refer to the sum of symptoms for a diagnostic variable or an individual’s match to a diagnostic prototype (Table 1, Level 1; Markon et al., 2011; Westen et al., 2010). These variables are dimensional rather than categorical in the sense that they are ordinal and continuous, meaning that people can vary from one another in terms of multiple degrees of severity of symptoms or similarity to a prototype. For example, a person who has 6 out of 9 symptoms is thought to be more severe

than a person with 3, or a person who is a 4/5 match to a prototype is a better candidate for diagnosis than a person who is a 2/5 match.

A second use of the term dimension refers to cases in which the indicators in these kinds of symptom counts fit factor models with a dominant superordinate or group factor, in either a unidimensional or hierarchical model (Table 1, Level 2). In this case, summing symptoms is justified by the fact that covariance analyses support considering them as a homogeneous group of indicators of the same underlying construct.

Table 1. *Three possible meanings of the term “dimensional” to describe psychological scales.*

Level	1	2	3
Features	Continuous	Continuous Homogeneous	Continuous Homogenous Distinct
Criterion	Indicators can be summed or averaged	Fit of indicators to a superordinate factor model	Structural distinction within a multidimensional framework and evidence of discriminant validity
Example	BDI HRSD	PROMIS depression TDI	NEO-PI-R depression PAI depression

Note. BDI = Beck Depression Inventory. HRSD = Hamilton Rating Scale for Depression. PROMIS = Patient-Reported Outcomes Measurement Information System. TDI = Teate Depression Inventory. PAI = Personality Assessment Inventory. NEO-PI-3 = Third edition of the NEO Personality Inventory.

In a third meaning, scholars have used the term “dimensional” to describe continuous constructs that are relatively homogenous and empirically distinct within the context of a multidimensional model (Table 1, Level 3; Waugh et al., 2017; Widiger, 2005). To meet this criterion, constructs need to be embedded in models that depict the covariance of multiple variables and be relatively circumscribed in construct breadth within that context. These

kinds of multidimensional models can be used to distinguish variables in terms of their breadth and similarity, both of which can be quantified and diagrammed. The most common types of multidimensional models come from multiscale assessment measures (Ben-Porath & Tellegen, 2020; Morey, 1991), or hierarchical models of personality or psychopathology generated through factor analysis (Kotov et al., 2017; McCrae et al., 2015).

Problems arise when people who use the word “dimension” to mean different things debate about how to advance the assessment of psychopathology. For example, Shedler et al. (2010) argued that personality disorders should be conceptualized as “syndromes” that reflect configurations of features that are notable in the psychiatric literature and are relatively easily recalled or formulated in the minds of clinicians (Spitzer et al., 2008). Because such syndromes can be conceptualized and measured in terms of similarity to diagnostic prototypes or symptom counts, Shedler and colleagues referred to them as “dimensions” (e.g., Shedler & Westen, 2007). Using a similar logic, some authors have argued that a simple solution to the problem of dimensionalizing personality disorders would be to compute symptom counts of traditional personality disorder categories (Millon, 1996; Oldham & Skodol, 2000; Zachar & First, 2015). For instance, Zachar and First (2015) argued that this approach would “help clinicians think about personality in terms of dimensional profiles rather than categories” and offered a profile sheet to facilitate this kind of assessment.

However, a large literature using a range of measurement tools has shown that the covariance of indicators of these syndromes do not fit unidimensional factor models (Nestadt et al., 2006; Sharp et al., 2015; Widiger & Simonsen, 2005; Wright et al., 2016). Moreover, multidimensional models can account for the variation in prototype or syndrome-based systems, such that different elements of the syndrome are related to different dimensions within a more comprehensive framework (Eaton et al., 2011; Mullins-Sweatt & Widiger, 2007). Moreover, such multidimensional models tend to explain the variance in criterion variables better than prototype or syndrome-based systems (Samuel et al., 2013; Yalch et al., 2012). For researchers who use the term dimension to describe variables within such

multidimensional frameworks, describing prototype scores or symptom counts as “dimensions” dilutes the meaning of the term and obscures the psychopathology assessment literature. From this perspective, symptom counts or prototype ratings do not reflect dimensions because they amalgamate a number of different, overlapping constructs, creating discriminant validity problems and disconnecting the variables from the general science on the structure of personality and psychopathology.

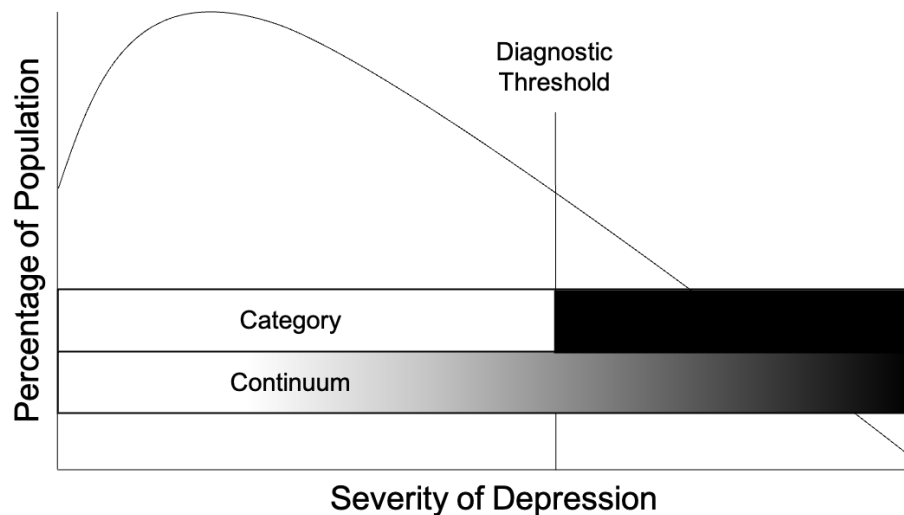
And yet, both of these groups say that they are advocating for “dimensional” models over categorical models. Some authors have even used the term dimensional to mean different things within the same paper (Huprich, 2018; Huprich & Bornstein, 2007; Morey et al., 2012; Simonsen, 2010; Spitzer et al., 2008). Scholars, clinicians, and students who do not or have not followed these debates closely would be understandably confused about what, exactly, the word dimension is supposed to mean. Thus, the importance of establishing clear and empirically anchored distinctions between possible meanings of this term, a goal to which we now turn.

“Dimensional” as Continuous

The first possible meaning of “dimension” is that the symptoms or indicators of a disorder category can be thought of or treated as continuous, in the sense of having additivity or ordinality properties (Table 1, Level 1). Practically speaking, the difference between dimensions as continua and categories is that the indicators of continua can be meaningfully summed or averaged, whereas the indicators of categories are used to make binary diagnostic decisions. For instance, depression is positively skewed in the population, such that most people have few symptoms, but some people have many symptoms (Figure 2). Given this distribution, patients can either be rated in terms of how many symptoms of depression they have (continuum), or they can be diagnosed as “having” depression or not (category). In either case, the more symptoms, the more likely the person would be described as depressed. Treating disorder categories as continuous variables increases their internal consistency and predictive validity relative to treating them as categories

(Markon et al., 2011). This has been one of the arguments in favor of a transition to “dimensional” diagnosis.

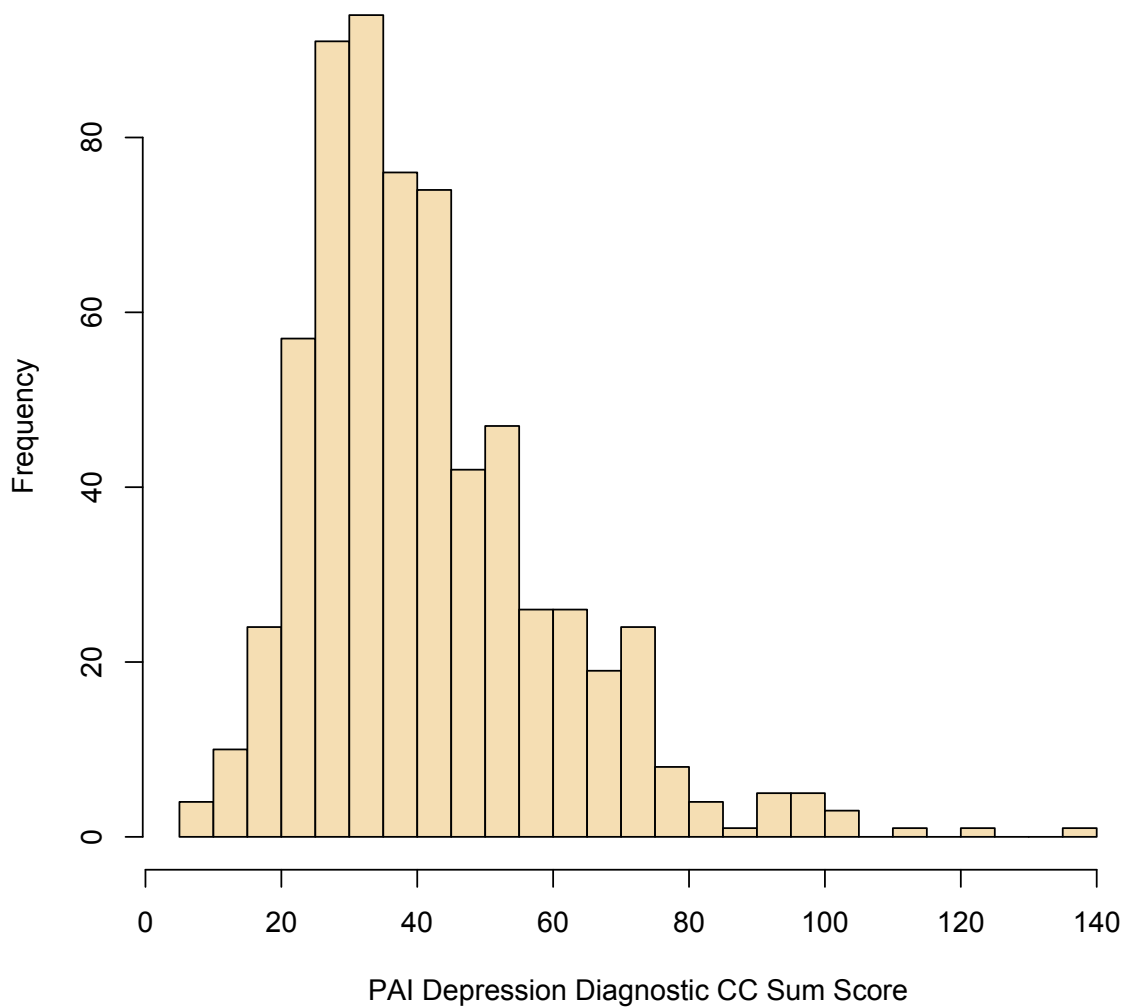
Figure 2. *Distribution of depression symptoms in the population.*



Note. This figure is expository and not based on real data. In the case of both a categorical and continuous variable, the diagnostic threshold is arbitrary. However, variance above and below the threshold is captured by the continuum and disregarded by the category.

Figure 3 illustrates this concept using the Personality Assessment Inventory Depression Diagnostic Consideration Cluster (PAI DDCC; Morey, 1996). The PAI DDCC comprises seven scales that map onto the diagnostic construct of Major Depressive Disorder as represented in the DSM: cognitive, affective, and somatic symptoms of depression, suicidality, concentration and thinking difficulties, social withdrawal, and low positive self-evaluation. These scales come from different constructs within the PAI (depression, suicidality, schizophrenia, and mania), but together they represent the symptoms within the diagnostic construct of Major Depressive Disorder and validly predict diagnostic status; as such they can be summed to create an overall depression diagnostic index of severity. As is evident in Figure 3, this sum score is continuously distributed in the normative sample ($N=1000$; Morey, 1991), and in this sense represents a depression dimension. The sum score appears very similar to a normal distribution, albeit more positively skewed as one would expect of a psychopathology scale.

Figure 3. *Distribution of Personality Assessment Inventory Depression Diagnostic Consideration Cluster (CC)Sum Score*



At the same time, the scales, although correlated with one another, are reasonably heterogeneous, in the sense that a single superordinate factor model does optimally account for their covariation. Parallel analysis suggests a 3-factor model of the subscale scores, with one factor reflecting the cognitive and affective components of depression, along with suicidality and social withdrawal; another factor reflecting concentration difficulties; and another reflecting low positive self-evaluation. RMSEA values for the 1, 2, and 3-factor

models were .13, .089, and .039, respectively, suggesting poor fit for all but the 3-factor model, and within the 3-factor model, the third factor was weakly related to the other two ($r=.19$ and $-.03$ with the first and second factor, respectively). In this way, although a score can be calculated that is roughly continuously distributed, it is nevertheless heterogeneous and not well-described by a single superordinate factor in a statistical sense. This kind of heterogeneity has also been observed in the polythetic DSM Major Depressive Disorder symptom set measured with other instruments (Kendler et al., 2013; van Loo et al., 2012). This heterogeneity has implications for explanation as well as prediction of external variables, in the sense that a single score might obscure differential relationships of the constituent elements.

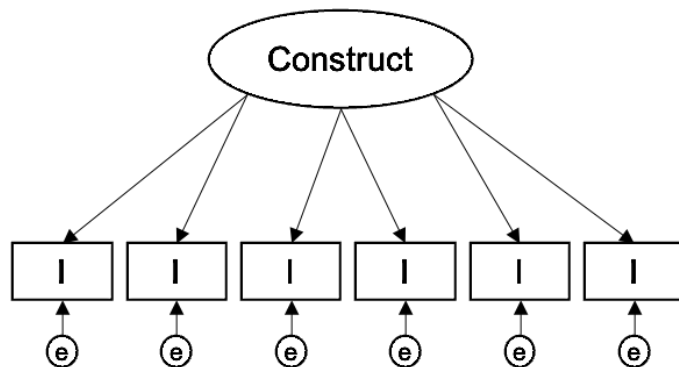
“Dimensional” as Homogeneous and Continuous

A different, and more restrictive, meaning of “dimension” is a variable for which there is structural (for example, factor analytic or latent variable) evidence that the indicators can be described in terms of a superordinate construct (Table 1, Level 2). That is, there is some manifold among all the putative indicators, they are internally covarying, and are homogenous to at least some extent (Figure 4). Evidence that a scale is unidimensional provides support for a parsimonious interpretation of the respondent’s total score.

Many tools that are commonly used by clinicians and psychological scientists (e.g., Beck et al., 1996; Hamilton, 1960) do not fit this description (Balsamo et al., 2014; Fried et al., 2016). For instance, the *Beck Depression Inventory* (BDI) is almost always treated as a unidimensional measure in practice, for which patients receive a total score that is interpreted as the severity of their depression (i.e., Table 1, Level 1). However, factor analytic research has suggested that the BDI has different factorial structures, depending on issues such as the population in which the measure is used, the version of the instrument being evaluated, and the analytic approach that is used to evaluate it (Campbell et al., 1984; Miles et al., 2001; Salamero et al., 1994; Toledano-Toledano & Contreras-Valdez, 2018; Whisman et al., 2000). The most common finding suggests correlated somatic and cognitive

factors, albeit with different item sets across studies. Although the correlation between these two factors is sufficiently high that some authors have proposed ignoring the difference between them (Brouwer et al., 2013; Quilty et al., 2010), covariation of BDI items never seems to fit a unidimensional model well.

Figure 4. *Unidimensional measurement model.*



Note. I = indicator. e = error. This model assumes that all indicators are continuous. In this model, all of the reliable covariance among indicators can be explained by the higher order factor. Any variance in indicators not explained by the higher order factor is specific to that indicator.

Contrast scales like the BDI with the *Patient-Reported Outcomes Measurement Information System* (PROMIS) depression scale, whose items were selected based on a series of psychometric analyses designed to ensure that the scale was unidimensional, meaning that a model for covariation among all indicators could be explained by a single higher order factor (i.e., Table 1, Level 2; see Irwin et al., 2010; Reeve et al., 2007). Confirmation that the PROMIS depression scale is unidimensional enabled further analyses in an item response framework to select items that were maximally discriminating across different levels of the depression continuum, and which operated similarly across demographic groups. Although other standalone unidimensional depression scales exist in the literature (e.g., Balsamo et al., 2014), they are not as common in practice as measures like the BDI that are treated as unidimensional despite empirical evidence that they are not.

“Dimensional” as Distinct, Homogenous, and Continuous

A third, and still more restrictive, possible definition of “dimension” is that a measure represents a single dimension that can be distinguished from “sibling” constructs – meaning constructs that measure similar albeit distinct aspects of psychopathology (Lawson & Robins, 2021; Table 1, Level 3). Distinct dimensions are not only continuous and unidimensional, they are also empirically differentiable from other similar constructs within a multidimensional framework. Distinctness is not a direct implication of the presence of dimensions, but it is one of the main interpretive advantages of identifying homogeneous dimensions. When such dimensions are distinct, they can be interpreted independently, and they have some potential for incremental value in making certain kinds of inferences.

Higher-order and hierarchical factor models provide a compelling framework for organizing variables in terms of their similarities and differences to one another based on their covariance and breadth (Brunner et al., 2012; Emmons, 1995; Markon, 2009; Markon et al., 2005). These models include broad factors influencing a relatively large number of observed variables, as well as narrow variables influencing a smaller number of observed variables. The scope of each broad factor can in this way be parsed into different, more specific units. For instance, a broad construct like neuroticism could be parsed into units involving anger and aggression and other units involving anxiety and worry. These models can have multiple levels, with lower-order variables being as narrow as individual indicators (Möttus et al., 2017) and higher-order variables explaining covariation among all variables in the system (Sharp et al., 2015) (Figure 5). The key feature of these models is that within each level, there is empirical evidence that the variables within each level capture covariation among different sets of indicators in the level below. At any given level within such a hierarchical model, the distinctiveness of “sibling variables” that are similar to one another in the sense that they represent aspects of some higher-order variable can be examined both by fitting factor models that test whether they represent different factors and by examining their discriminant validity with respect to external variables, course, and other considerations (Lawson & Robins, 2021).

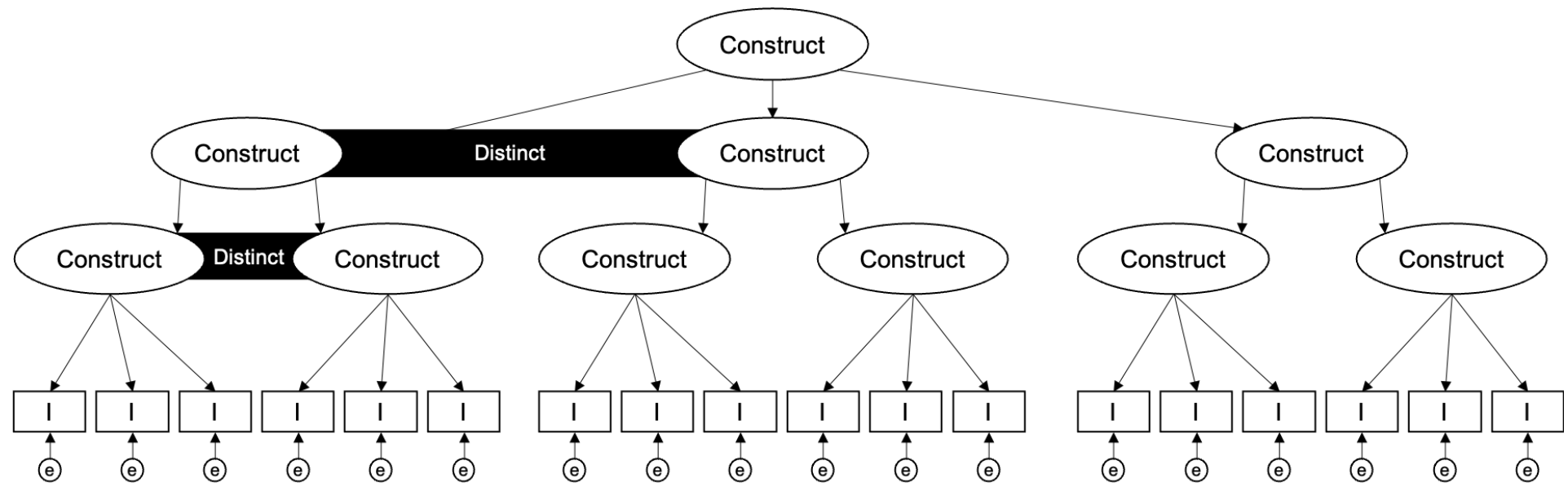
Such models have been common in trait psychology for decades. For instance, the NEO-PI (McCrae et al., 2005), one of the most commonly used personality trait measures, has five higher order domains, each of which has six lower order facet scales. One of the facets of the neuroticism domain is depression. Depression is continuous and homogeneous as a scale, and it is also empirically distinguishable from other facets within that domain such as anxiety or angry hostility in factor models of the instrument and in external validity analyses. For instance, Samuel and Widiger (2006) showed that the depression scale was more strongly related than angry hostility to measures of dependency but was more weakly related than angry hostility to measures of narcissistic and antisocial personality.

Hierarchical models are also common in maladaptive trait systems. For example, the *Personality Inventory for DSM-5* (PID-5; Krueger et al., 2012) offers a trait approach to accounting for personality disorder symptoms. It has a depressivity scale that cross-loads on a higher order detachment and negative affectivity factors, and it is empirically distinct from other scales that load onto those factors, such as withdrawal (detachment) and anxiousness (negative affectivity).

Like these measures, the PAI also uses a hierarchical logic for distinguishing higher-order from lower-order constructs and constructs within the same level from one another. However, the PAI differs from the NEO-PI or PID-5 in that PAI depression is a higher order construct with lower-order scales, whereas in the NEO-PI or PID-5 depression is a lower order facet of a higher-order trait domain. The PAI depression scale is conceptualized as a broad construct with three underlying (cognitive, affective, and physiological) factors, each of which is measured with 8 items. In contrast to the PAI DDCC described above, this hierarchical structure, in which 8 items load onto 3 distinct factors, which in turn load onto a higher order depression factor fit the data well in the normative sample ($CFI = .98$; Morey, 1991). The subscales are also distinct in relating to different criterion variables. For instance, the physiological depression scale is relatively sensitive to the diagnosis of PTSD (McDevitt-Murphy et al., 2005) whereas the affective depression scale is uniquely related to a performance-based measure of emotional distress (Petrosky, 2004).

Thus, the PAI offers two approaches to conceptualizing depression that vary in the degree to which they can be considered “dimensional”. One, the PAI DDCC, is designed to mirror the symptom set of the DSM Major Depressive Disorder construct. While it is continuous and its constituent scales are both homogeneous and distinct in terms of external validity, its scales do not cohere as a valid superordinate construct. The other, the PAI Depression scale, meets all the criteria described in Table 1. The domain and each of the subscales can be summed as continuous measures, each of its subscales are homogeneous, together they form a valid superordinate construct, and each is distinct in terms of both internal and external validity.

Figure 5. *Higher-order factor model.*



Note. I = indicator. e = error. This model assumes that all indicators are continuous. Each construct is unidimensional, in that all of the covariance among indicators can be explained by the higher order factor and any variance in indicators not explained by the higher order factor is specific to that indicator. Within each level of the hierarchy, each construct is also distinct.

Information Value of Dimensions

Up until this point, we have described different conceptualizations of a “dimension” but have not articulated the value or theoretical role of these different levels of dimensionality, in the sense of how they might function in different structural and measurement paradigms. For instance, a traditional assumption is that the covariation of indicators on a measure can be explained by a hypothetical *reflective* construct. However, *formative* constructs involving uncorrelated indicators (Blalock, 1971; Bollen & Lennox, 1991) have also been proposed, as have *mutualist* or *network* structures that comprise mutual or dynamic relationships directly between indicators (Figure 6; Boorsboom & Cramer, 2013; Cramer et al., 2010; Schmittman et al., 2013). It is natural to ask what roles the levels of dimensionality are playing in these different paradigms. Our suggestion is that the different levels of dimensionality (Table 1) constitute levels by which a measure instantiating the dimension (that is, a sum score or trait estimate) provides fidelitous, unique summary information about the set of indicators. As one moves to lower levels, there is greater ambiguity about the information conveyed by the indicators in a measure; as one moves to higher levels, the measure represents more unique and specific information about the indicators as a set.

As we have articulated in earlier papers (Jonas & Markon, 2016; Markon & Jonas, 2016; Markon, in press) a latent variable can be understood as a representation of a set of observed indicators, with a measure of that latent variable for a respondent (such as a trait estimate or sum score) representing that individual's observed responses. The measurement process in this way is a reencoding or mapping of a set of observed responses into a value (for example, a score or latent variable estimate) that summarizes those responses. To the extent that a set of observed variables is highly correlated, and thus fits a superordinate factor model well, the representation implies less loss of information. In the extreme case, if all the observed variables are perfectly correlated with one another, there will be no loss of information in expressing the observed variables with a single latent variable. Similarly, if one set of observed variables is highly correlated within the set, but

uncorrelated with variables of another set, it implies latent variables representing each set are uniquely informative about their corresponding set, with no loss of information about the “source” of the set due to cross-representations associated with cross-correlations (cf., Markon, in press).

Seen from this perspective, the levels of dimensionality in Table 1 imply different degrees of fidelity of representation by the corresponding latent variables. In a classical account of latent variables, an exemplar at the first level (i.e., continuous variable) with uncorrelated indicators, implies a latent variable with significant loss of information in summarizing the observed responses with a single score. The second level (i.e., homogeneous dimension) implies a latent variable with greater fidelity in representing the observed responses, but possible loss of information due to which subsets of indicators were “driving” the summary. The third level (i.e., distinct dimension) represents the case where indicators can be represented by a superordinate latent variable with substantial fidelity, and no loss of information about the sources of that information, because the responses are “informationally pure” in an important sense. These distinctions parallel concerns that have been raised about heterogeneity in some personality disorder constructs, such as when a diagnostic criteria can be met in multiple ways (Clarkin et al., 1983): even if the diagnosis has external validity, a group of individuals meeting the criteria might nonetheless differ in ways that are important for etiologic or treatment reasons.

However, distinctions between reflective versus formative latent variable accounts may be instructive in highlighting different functions that the various levels might serve. Reflective accounts — the classical accounts of latent variables — can be distinguished from formative accounts by four criteria (Engelhard & Wang, 2014; Jarvis et al., 2003; Willoughby et al., 2014). First, with reflective variables, the latent construct causes variation in the indicators, whereas in formative variable, the causal path is assumed to go in the other direction. An example of a formative construct is socioeconomic status (SES), for which education, inherited wealth, or income could all increase SES, but we do not typically think of an increase in a latent SES variable as having a causal effect on education, inherited wealth,

and income. Second, the indicators of a reflective variable are sampled from a broader population of possible markers, and are thus exchangeable, whereas the indicators of a formative construct are all necessary to conceptualize that construct, and thus are not exchangeable. To continue with the example, if income were removed from the indicator list of SES, SES would become a qualitatively different variable. Third, the indicators of reflective variables should covary, and that covariance should be explained by the underlying latent construct, whereas the covariance of indicators is irrelevant to formative variables. It is the case that education, wealth, and income are likely to covary, but this is not important for calculating SES. Fourth, each indicator of a reflective variable should have a similar pattern of covariation with external outcomes, whereas this is not necessary for formative variables. One could imagine certain correlates of income that do not correlate with inherited wealth (e.g., conscientiousness).

Without the criterion of indicator covariance, dimensions in a formative account imply potential loss of information in summarizing indicators by the dimension (Level 1). When formative indicators are uncorrelated with one another, the formative dimension is acting as a proxy summary of a set of heterogeneous variables, with potentially substantial loss of information. For example, if one summarizes a set of uncorrelated socioeconomic variables in terms of a single SES dimension, substantial unique predictive and other information carried by each of the indicators is lost in creating the summary dimension. This is consistent with other characteristics of formative accounts, such as different indicators having different patterns of covariation with external outcomes. It is also consistent with exchangeability-related critiques of variables often assumed to be formative, such as SES, insofar as the meaning of a formative variable can change from measure to measure, as can its predictive characteristics (e.g., Antonoplis, 2022). Similar arguments about heterogeneity have been made in the domain of polythetic personality disorder diagnostic systems (Burger, Jessen, & Freyberger, 2000; Cooper, Balsis, & Zimmerman, 2010).

At the same time, is important to note that such formative dimensions may have external validity outside of internal structural considerations. For instance, risk estimates for

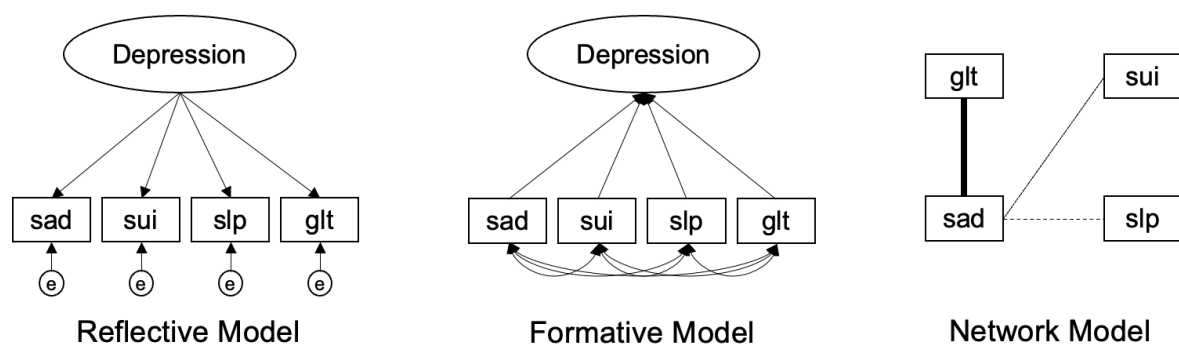
psychopathology-relevant outcomes such as suicide or violence often involve a summation of a set of uncorrelated observed variables. While this risk is conceptualized continuously, there is no presumption that some latent construct is “causing” the variation in the observed variables. Chronic health problems and lack of social support both are observable indicators of suicide risk, but for that purpose there is no need to presume that they stem from a common source. However, the aggregate index does not summarize well, in an information-theoretic sense, the observed values of the indicators, in that each of these indicators includes considerable information that is independent of the summary measure. Thus, although the information-theoretic summary value of a dimension increases with the levels in Table 1, a dimension at lower levels may have other, external-predictive value.

Mutualist or network-based accounts are another measurement paradigm that these levels of dimensionality might be applied to. In this paradigm, symptoms reflect behaviors or neuropsychological states that directly influence one another within associative or causal networks (Boorsboom & Cramer, 2013; Cramer et al., 2010; Schmittman et al., 2013). The network approach assumes that it is the symptoms themselves that are of primary causal interest. For example, guilt might cause sadness, which causes suicidality and sleep problems; the direction and strength of direct connections between symptoms can be evaluated, as has been done in several studies of depression (Bringmann et al., 2015; Murri et al., 2020; von Borkulo et al., 2015). From this perspective, it is often assumed that conceptualizing symptoms in terms of causal latent entities obscures their interconnections. Although reflective models seem to fit naturally within a medical model of psychopathology, in which an underlying syndrome (depression) causes symptom expression (sadness, guilt, suicidality, sleep problems), in medicine, it is often possible to separate the disease from the symptoms. For instance, the reason why early detection of cancer via regular screening is so important is that it is possible for a person to have cancer without symptoms (Boorsboom & Cramer, 2013).

Dimensionality is a relevant consideration within these mutualist or network accounts as well. Latent variable and network models can be treated as equivalent within a common

graph modeling framework. Within network accounts, there is often a need to account for superordinate patterns of relationships among variables; these are referred to in different ways, such as communities or clusters, but they are a form of latent variable. For instance, within psychopathology networks, one might find that guilt, shame, and sadness are more highly related to one another than they are to alcohol use, heroin use, and dangerous behavior, which are in turn highly related with one another. Regardless of the causal nature of direct paths between the variables, dimensions such as internalizing and externalizing might be invoked to account for the clustering of these paths. As in traditional latent variable models, the dimensions are being used to summarize the subordinate observed variables, with loss of information in measurement using those dimensions being proportional to their strength and patterns of covariance between the indicators.

Figure 6. *Reflective, formative, and network models of depression.*



Note. sad = sadness. sui = suicidality. slp = sleep. glt = guilt. Arrows depict assumed causal directions in reflective and formative models. In network model, dashed line indicates a negative correlation and solid line indicates positive association, and bold line indicates stronger correlation. This figure is expository and not based on real data.

Additional Issues

Thus far, we have described how the term “dimension” has been used to refer to three distinct concepts in psychopathology and have clarified empirical criteria by which

those concepts can be distinguished. This discussion raises a few adjacent issues that we address in this section.

Internal Categorical Structure and Taxonicity

A typically implicit assumption of dimensional models is that they meet taxometric evidence that the symptoms are distributed continuously in the population. Taxometric studies apply “consistency tests” to indicators of a construct that would be satisfied only if a variable represented a taxon (i.e., category) (Meehl, 1995; 1999). This approach is applied in psychopathology research because of the absence of any definitive indicator that underlying psychopathology is “present” or “absent” (e.g., lab test that shows the presence of cancer cells), and thus the necessity of making that determination within the pattern of data themselves. A variety of taxometric methods are available based on different assumptions about the nature of taxa and the quality of indicators (Ruscio et al., 2013). What they generally have in common is that certain patterns of symptom covariance will tend to be more likely if a variable is a taxon than if it is not.

Empirical evidence from taxometric studies suggests that depression symptoms (Ruscio & Ruscio, 2000), like the symptoms of most other forms of psychopathology (Haslam et al., 2020), are not taxonic or categorical, but rather are continuous. Strictly speaking, this kind of evidence should be considered a prerequisite to describing a variable as “dimensional” across all three levels in Table 1, although one could use a continuous observed measure as an indicator of taxon membership. Conversely, even if there is evidence that a given construct is a taxon, it could nevertheless be rated with continuous severity scores. For instance, nationality is a taxon, such that a person could either be a citizen of the Netherlands or Belgium; but a Dutch person who lives near the border of Belgium could be said to be more Belgian than a Dutch person who lives in Friesland. Or, a Dutch-speaking Belgian could be scored as more Dutch than a French-speaking Belgian. Even though taxa could be represented with continuous scores that depict the probability of class membership, it would be illogical to describe them as dimensions or continuous variables.

Diagnostic Thresholds and their Internal and External Validity

Diagnostic thresholds function as a categorizing operation with regard to dimensions, in that they take a dimensional or continuous variable (such as a symptom count) and transform it into a categorical variable, typically in an effort to define “caseness”. If a form of psychopathology were taxonic, a reasonable practice would be to set a threshold on a continuous indicator such that the dimensional indicator is “cut” to produce most likely taxon classifications of respondents (for example, disordered versus not). Relatedly, distributional properties of the dimension that are multimodal could provide natural thresholds, such as a pronounced second mode found at an extreme. An example here can be found with the PAI Infrequency (INF) scale, a validity scale used to detect problematic responding, where the generally positive skew of this scale is often accompanied by a distinct secondary mode in samples with limited motivation (such as undergraduate samples). Here, the point of rarity between this secondary mode and the rest of the population provides a natural break point for establishing test validity.

Even if a form of psychopathology is not taxonic, it is nonetheless possible that a natural threshold on a dimension may exist with which to provide meaningful categorization (Blanton & Jaccard, 2006; Jonas & Markon, 2013; Markon, Chmielewski, & Miller 2011; Morey, 2021). Some dimensions may demonstrate potential thresholds where there is a nonlinear change in expected outcomes or antecedents. For example, the continuous indicators of HbA1c (in diabetes) and diastolic blood pressure (in hypertension) demonstrate ranges where mortality risk increases exponentially, providing a meaningful basis for establishing a threshold.

However, to date there is little evidence to support the existence of such meaningful thresholds for dimensions in psychopathology. Although the question of nonlinearity of associations with psychopathology has received less research attention than is perhaps merited, when it has been examined explicitly, it has generally found linear rather than nonlinear relationships with major functioning variables (Jonas & Markon, 2013; Markon, 2010). Similarly, research on most constructs using representative community samples

rarely finds discontinuities in these distributions. To the extent these findings would generalize to other variables, this suggests that there is no point along psychopathology continua to place diagnostic thresholds to statistically minimize loss of information in the prediction of important outcome variables. Barring such natural thresholds, there is little non-arbitrary rationale for choosing one point or another to determine caseness. Although conventions have evolved invoking distributional properties, such as being 1.5 (e.g., learning disorder) or 2.0 (e.g., intellectual disability) standard deviations from the population mean, even a more widespread application of these conventions would result in similar prevalence of “cases” across constructs with distinctly different implications for functioning and need for intervention.

As such, cut scores have typically been decided by diagnostic manual committees based on relatively simple rules (e.g., more than half of all possible symptoms). For the Alternative Model of Personality Disorder in the DSM-5, a fundamentally dimensional diagnostic system, the diagnostic threshold for personality disorder “caseness” was explicitly calibrated to parallel the (largely arbitrary) precedent established in previous editions of the manual, with the goal of providing continuity for public health systems and for the research literature (Morey & Skodol, 2013), while emphasizing the continuity of the phenomenon on either side of this threshold.

It is worth noting that the provision of a clear cut-off for diagnosis may have masked the arbitrariness of diagnostic thresholds in a way that is exposed to a greater degree by continuous variables (Krueger et al., 2014). Nonetheless, deciding how severe a person’s problems should be for diagnosis is obviously important for practical and public health reasons. Absent natural thresholds, this determination becomes an ethical or sociopolitical question, having to do with factors such as the level of concern a society has with mental health problems, considerations regarding the balance between providing care and stigmatizing human variability, the resources a society has to help people with mental health difficulties, and the value placed on treating every person the same way.

Context

Our goal has been to demarcate three distinct possible meanings of the word “dimension” as it applies to psychopathology, in the hopes that these meanings will be less entangled moving forward. However, we did not intend to imply that any single meaning is necessarily more important than the others, or that a particular psychopathology variable or measure will meet a certain dimensionality criterion across all contexts. There may be contexts in which different kinds of dimensional evidence is most important. There may even be contexts in which you would not want to achieve all aspects of dimensions. For instance, in a screening setting, the structural integrity of a measurement model may be less important than a general indication of mental health symptoms of any kind. Level 1 may be the most important characteristic of a mental health screening tool in some settings, and thus a formative dimension may be preferred.

Our main point is that our measurement tools, language, and evidence should align with one another to avoid confusion and facilitate science-practice integration. When considering theoretical latent psychopathology variables, it is typically possible to generate measures of most psychopathology constructs that are dimensional in all three possible senses articulated above. As such, we do think clinical tools designed to measure such constructs should also generally meet all three standards, with some possible exceptions, such as the screening situation described above. However, we remain open to the possibility that some forms of psychopathology may be taxa, may be described by valid but non-overlapping indicators (Loevinger, 1954), or may otherwise not reflect dimensional latent constructs.

Another type of context is measurement approach. Examples in this review were limited by the use of a single approach to measurement and research design, namely questionnaires as implemented in cross-sectional data. It is possible that different kinds of psychometric properties would emerge for the same psychopathology variables using different measurement methods, and there are reasons to be sanguine about the importance of multimethod data to help us better understand the nature of psychopathology. Similarly, it is possible that measurement properties may not generalize across populations. The general

need to sample more broadly in psychological research is well-known (Thalmayer et al., 2021).

Summary

The goal of this paper was to distinguish three meanings of the word “dimension” as it has been applied to debates about the nature of psychopathology. The first meaning is that indicators can be summed to form a continuous score. The second meaning is that the indicators fit a unidimensional measurement model that constrains their covariance to a single higher order dimension. The third meaning is that the variable is situated in a multivariate system within which different dimensions can be empirically distinguished from one another. Distinguishing these meanings carefully is important for engaging in a more coherent discussion regarding how to best transition to more evidence-based models of psychopathology in research and practice.

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